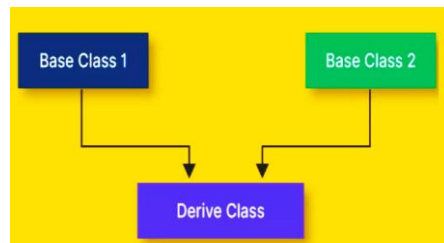


5-1 Multiple INHERITANCE

Multiple Inheritance is a feature of C++ where a class can inherit from more than one base class. It allows a derived class to combine properties and behavior from multiple parent classes, making code more reusable and flexible.

A class can be derived from more than one base class. For example:

- A CHILD class is derived from the FATHER and MOTHER class



Syntax:

```
class A
{
// Base class A };
class B
{
// Base class B };
class C : public A, public B
{
// Derived class };
```

Example (1):

The following example demonstrates how a class can inherit from multiple base classes:

```
#include <iostream>
using namespace std;

class A {
public:
    A() { cout << "Class A \n"; }
};
```

```

class B {
public:
    B() { cout << "Class B \n"; }
};

// Inheriting from both class A & B
class C : public B, public A {
public:
    C() { cout << "Class C \n"; }
};

int main() {
    C obj;
}

```

Output:
Class B
Class A
Class C

Example (2):

Parametrize the constructor and override function with inheritance.

```

#include <iostream>
using namespace std;
// multiple inheritance
class A {
private:
    string name;
    int age;
public:
    A(string n, int a)
    {name =n;
    age = a; }
    void print(){
        cout<<"your name is "<< name<<endl;
        cout<<"your age is "<< age << endl;
    }
};

class B {
private:
    string email;
public:
    B(string em) {
        email=em; }
    void print(){
        cout<<"your email is "<< email<<endl;
    }
};

// Inheriting from both class A & B

```

Output:
I the constructor from C class
your name is Ahmed
your age is 20
your email is ahmed@gmail.com
grade final is 85

```

class C : public B, public A {
private:
    int gradefinal;
public:
    C(string n, int a, string em, int gf) : A(n,a), B(em)
    {
        cout<<"I the constructor from C class \n";
        gradefinal = gf;
    }
    void print(){
        A::print();
        B::print();
        cout <<"gradefinal is "<< gradefinal<<endl;
    }
};

int main() {
    C obj("Ahmed", 20, "ahmed@gmail.com", 85);
    obj.print();
}

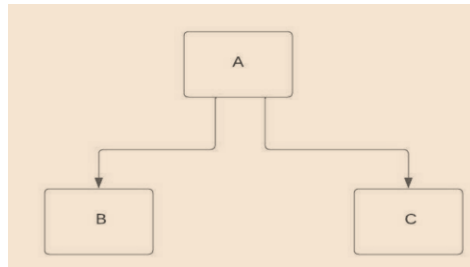
```

Parametrize the constructors

override

5-2 Hierarchical Inheritance

In hierarchical inheritance, more than one subclass is inherited from a single base class. i.e. more than one derived class is created from a single base class. For example, cars and buses both are vehicle.



Example:

```

#include <iostream>
using namespace std;

class Vehicle
{
public:
    Vehicle()
    {
        cout << "This is a Vehicle" << endl;
    }
};

```

```
class Car : public Vehicle
{
public:
    Car()
    {
        cout << "This Vehicle is Car" << endl;
    }
};

class Bus : public Vehicle
{
public:
    Bus()
    {
        cout << "This Vehicle is Bus" << endl;
    }
};

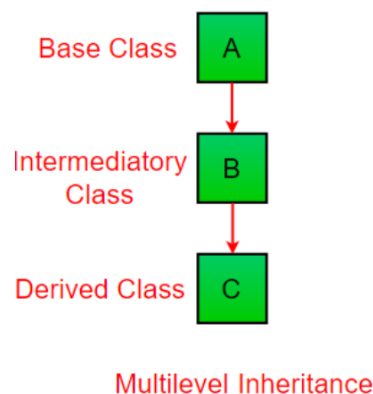
int main()
{
    Car obj1;
    Bus obj2;
    return 0;
}
```

Output:

This is a Vehicle
This Vehicle is Car
This is a Vehicle
This Vehicle is Bus

5- 3. Multilevel Inheritance

Multilevel inheritance in C++ means a class is derived from another derived class, forming a chain of inheritance.



Example :

Transitive Nature The most derived class inherits all the features of all the classes in the hierarchy, but whether they are accessible depends on the access specifier.

```
#include <bits/stdc++.h>
using namespace std;

class Grandfather {
public:
    void g() {
        cout << "This is a Grandfather class" << endl;
    }
};

class Father : public Grandfather {
public:
    void f() {
        cout << "This is Father class" << endl;
    }
};

class Child : public Father {
public:
    void c() {
        cout << "This is Son class" << endl;
    }
};

int main() {

    // Creating object of Child class
    Child obj;

    // Calling member methods of all classes
    // using Child object
    obj.c();
    obj.Father::f();
    obj.Father::Grandfather::g();
    return 0;
}
```

Output

```
This is Son class
This is Father class
This is a Grandfather class
```

H.W: change the access specifier to private and note what is happen to the output?

H.W.: what is the Diamond Problem in inheritance?

H.W. : how can we build Hybrid Inheritance program In C++.

H.W. search about Destructor Order in inheritance programs in C++.